

Internet Appendix for “Integrating Factor Models”

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ABSTRACT

This Internet Appendix provides additional tables and figures supporting the main text.

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I. Conditional Asset Pricing Models

I.A. Marginal Likelihood for the Unrestricted Case

The multivariate representation of the beta pricing equations (1) and (2) is given by

$$\begin{aligned} R &= \iota_T \alpha'_0 + Z_{-1} \alpha'_1 + F \beta'_0 + \Xi \beta'_1 + U_R \\ &= W \Phi + U_R \end{aligned} \tag{IA.1}$$

$$F = X A_F + U_F, \tag{IA.2}$$

where $R = [r_1, \dots, r_T]'$, $F = [f_1, \dots, f_T]'$, $X = [\iota_T, Z_{-1}]$, $Z_{-1} = [z_0, \dots, z_{T-1}]'$, ι_T is a T -vector of ones, $\Xi = [\xi_1, \dots, \xi_T]' \in \mathbb{R}^{T \times (km)}$ is defined through $\xi_t = (I_k \otimes z_{t-1}) f_t$, $W = [X, F, \Xi]$, $\Phi = [\alpha_0, \alpha_1, \beta_0, \beta_1]'$, $U_R \in \mathbb{R}^{T \times (N+K-k)}$ and $U_F \in \mathbb{R}^{T \times k}$ are the matrices of residuals, N is the number of test assets, K is the total number of factors, and $K - k$ is the number of factors that are added to the test assets on the left-hand side of equation (IA.1). The k remaining factors are on the right-hand side of equation (IA.1) and the left-hand side of equation (IA.2). The number of retained predictors is denoted by m , and it ranges, in the conditional setting, between one and M . In the unconditional setting, the number of retained predictors is equal to zero everywhere.

We assume that the error terms $U_R = [u_{r,1}, \dots, u_{r,T}]'$ and $U_F = [u_{f,1}, \dots, u_{f,T}]'$ conditional on the parameter space obey the normal distribu-

tion

$$\begin{pmatrix} u_{r,t} \\ u_{f,t} \end{pmatrix} \sim \text{i.i.d. } f_{\text{MV}}(0, \Psi), \text{ where } \Psi = \begin{pmatrix} \Sigma_{RR} & 0 \\ 0 & \Sigma_{FF} \end{pmatrix}. \quad (\text{IA.3})$$

The factor and return innovations are orthogonal through equation (IA.1). While the conditional distribution of the error terms is normal, the unconditional distribution (upon integrating out the parameter space) essentially departs from normality. Tu and Zhou (2004) show that certainty equivalent losses associated with ignoring fat tails in the conditional distribution are small, suggesting that the conditional normality assumption could well characterize the stock return distribution from a decision-making perspective. Then, based on the multivariate representation of the beta pricing equations, the likelihood of the hypothetical sample can be written as

$$\begin{aligned} \mathcal{L}(D_0|Z_0, \Sigma_{RR}, \Sigma_{FF}, \Phi, A_F, \mathcal{M}) \propto \\ |\Sigma_{RR}|^{-\frac{1}{2}T_0} \exp \left\{ -\frac{1}{2} \text{Tr} \left[\Sigma_{RR}^{-1} \left(R_0' Q_{W_0} R_0 + (\Phi - \hat{\Phi}_0)' W_0' W_0 (\Phi - \hat{\Phi}_0) \right) \right] \right\} \times \\ |\Sigma_{FF}|^{-\frac{1}{2}T_0} \exp \left\{ -\frac{1}{2} \text{Tr} \left[\Sigma_{FF}^{-1} \left(F_0' Q_{X_0} F_0 + (A_F - \hat{A}_{F_0})' X_0' X_0 (A_F - \hat{A}_{F_0}) \right) \right] \right\}, \end{aligned} \quad (\text{IA.4})$$

where the zero subscripts denote data (returns, factors, and predictors) from the hypothetical sample, and where

$$\hat{\Phi}_0 = \left[\mathbf{0}_{(N+K-k) \times (m+1)}, \hat{\beta}_0, \mathbf{0}_{(N+K-k) \times km} \right]' \quad (\text{IA.5})$$

$$\hat{A}_{F_0} = \left[\hat{f}, \mathbf{0}_{k \times m} \right]'. \quad (\text{IA.6})$$

Here, $\hat{\beta}_0 = (F'F)^{-1} F'R$ is the slope coefficient in a zero-intercept regression of the returns r_t on the factors f_t , $\mathbf{0}_{i \times j}$ is a matrix of zeros with i rows and j columns, and the operator $Q_J = I_T - J(J'J)^{-1}J'$ is defined for a matrix J with full column-rank and T rows. The hypothetical sample is weighted against mispricing and against time-varying moments.

Combining the likelihood of the hypothetical sample with the uninformative prior $\pi(\Sigma_{RR}, \Sigma_{FF}, \Phi, A_F) \propto |\Psi|^{-\frac{N+K+1}{2}}$ yields the informed prior, formulated as

$$\text{vec}(A_F) | \Sigma_{FF}, D_0 \sim f_{\text{MV}} \left(\text{vec}(\hat{A}_{F_0}), \Sigma_{FF} \otimes (X_0'X_0)^{-1} \right) \quad (\text{IA.7})$$

$$\Sigma_{FF} | D_0 \sim f_{\text{IW}}^k(F_0'Q_{X_0}F_0, T_0 + N + K - k - m - 1) \quad (\text{IA.8})$$

$$\text{vec}(\Phi) | \Sigma_{RR}, D_0 \sim f_{\text{MV}} \left(\text{vec}(\hat{\Phi}_0), \Sigma_{RR} \otimes (W_0'W_0)^{-1} \right) \quad (\text{IA.9})$$

$$\Sigma_{RR} | D_0 \sim f_{\text{IW}}^{N+K-k}(R_0'Q_{W_0}R_0, T_0 - (k+1)m - 1). \quad (\text{IA.10})$$

All quantities based on the hypothetical sample in equations (IA.7) to (IA.10) can be expressed in terms of quantities observed from the actual sample. Specifically, note that

$$X_0'X_0 = \frac{T_0}{T} (X'X) = T_0 \begin{pmatrix} 1 & \bar{z}' \\ \bar{z} & \hat{V}_z + \bar{z}\bar{z}' \end{pmatrix} \quad (\text{IA.11})$$

$$\begin{aligned} F_0'Q_{X_0}F_0 &= (F_0 - X_0\hat{A}_{F_0})' (F_0 - X_0\hat{A}_{F_0}) \\ &= (F_0 - \iota_{T_0}\bar{f}')' (F_0 - \iota_{T_0}\bar{f}') = T_0\hat{V}_f \end{aligned} \quad (\text{IA.12})$$

$$\begin{aligned} R_0'Q_{W_0}R_0 &= (R_0 - W_0\hat{\Phi}_0)' (R_0 - W_0\hat{\Phi}_0) \\ &= \frac{T_0}{T} (R'R - \hat{\Phi}_0'W'W\hat{\Phi}_0) = \frac{T_0}{T} (R'R - \hat{\beta}_0'F'F\hat{\beta}_0). \end{aligned} \quad (\text{IA.13})$$

We next express $W'_0 W_0$ in terms of quantities observed from the actual sample. First note that

$$X'_0 F_0 = (X'_0 X_0) \hat{A}_{F_0} = T_0 \begin{pmatrix} 1 & \bar{z}' \\ \bar{z} & \hat{V}_z + \bar{z} \bar{z}' \end{pmatrix} \begin{pmatrix} \bar{f}' \\ \mathbf{0}_{m \times k} \end{pmatrix} = T_0 \begin{pmatrix} \bar{f}' \\ \bar{z} \bar{f}' \end{pmatrix}. \quad (\text{IA.14})$$

We exploit the prior independence between F_0 and Z_0 to compute $Z'_0 \Xi_0$, $F'_0 \Xi_0$ and $\Xi'_0 \Xi_0$ as

$$Z'_0 \Xi_0 = \begin{pmatrix} \sum_{t=0}^{T_0-1} (z_{0,t}^1 \cdot (f'_{0,t+1} \otimes z'_{0,t})) \\ \vdots \\ \sum_{t=0}^{T_0-1} (z_{0,t}^m \cdot (f'_{0,t+1} \otimes z'_{0,t})) \end{pmatrix} = T_0 \begin{pmatrix} \bar{f}' \otimes \overline{z^1 \cdot z'} \\ \vdots \\ \bar{f}' \otimes \overline{z^m \cdot z'} \end{pmatrix} = T_0 (\bar{f}' \otimes \Gamma) \quad (\text{IA.15})$$

$$F'_0 \Xi_0 = \begin{pmatrix} \sum_{t=1}^{T_0} (f_{0,t}^1 (f'_{0,t} \otimes z'_{0,t-1})) \\ \vdots \\ \sum_{t=1}^{T_0} (f_{0,t}^k (f'_{0,t} \otimes z'_{0,t-1})) \end{pmatrix} = T_0 \begin{pmatrix} \overline{f^1 \cdot f'} \otimes \bar{z}' \\ \vdots \\ \overline{f^k \cdot f'} \otimes \bar{z}' \end{pmatrix} = T_0 (\Delta \otimes \bar{z}') \quad (\text{IA.16})$$

$$\text{where } \overline{z^i \cdot z'} = \frac{1}{T} \sum_{t=0}^{T-1} (z_t^i \cdot z'_t), \quad \overline{f^i \cdot f'} = \frac{1}{T} \sum_{t=1}^T (f_t^i \cdot f'_t), \quad \Gamma = \begin{pmatrix} \overline{z^1 \cdot z'} \\ \vdots \\ \overline{z^m \cdot z'} \end{pmatrix}, \text{ and}$$

$$\Delta = \begin{pmatrix} \overline{f^1 \cdot f'} \\ \vdots \\ \overline{f^k \cdot f'} \end{pmatrix}. \text{ Moreover,}$$

$$\Xi'_0 \Xi_0 = \sum_{t=1}^{T_0} (f_t \otimes z_{t-1}) \cdot (f'_t \otimes z'_{t-1}) = \sum_{t=1}^{T_0} (f_t f'_t \otimes z_{t-1} z'_{t-1}) = T_0 (\overline{f f'} \otimes \overline{z z'}) \quad . \quad (\text{IA.17})$$

It then follows that

$$\begin{aligned} W'_0 W_0 &= \begin{pmatrix} \iota'_{T_0} \iota_{T_0} & \iota'_{T_0} Z_0 & \iota'_{T_0} F_0 & \iota'_{T_0} \Xi_0 \\ Z'_0 \iota_{T_0} & Z'_0 Z_0 & Z'_0 F_0 & Z'_0 \Xi_0 \\ F'_0 \iota_{T_0} & F'_0 Z_0 & F'_0 F_0 & F'_0 \Xi_0 \\ \Xi'_0 \iota_{T_0} & \Xi'_0 Z_0 & \Xi'_0 F_0 & \Xi'_0 \Xi_0 \end{pmatrix} \\ &= T_0 \begin{pmatrix} 1 & \bar{z}' & \bar{f}' & \bar{\Xi}' \\ \bar{z} & \hat{V}_z + \bar{z} \bar{z}' & \bar{z} \bar{f}' & \bar{f}' \otimes \Gamma \\ \bar{f} & \bar{f} \bar{z}' & \hat{V}_f + \bar{f} \bar{f}' & \Delta \otimes \bar{z}' \\ \bar{\Xi} & \bar{f} \otimes \Gamma' & \Delta' \otimes \bar{z} & (\hat{V}_f + \bar{f} \bar{f}') \otimes (\hat{V}_z + \bar{z} \bar{z}') \end{pmatrix}. \quad (\text{IA.18}) \end{aligned}$$

Combining the likelihood of the observed sample $D = (R, F, Z)$ with the prior distributions based on the hypothetical sample (equations (IA.7) to

(IA.10)) yields the following posterior distribution:

$$\text{vec}(A_F) | \Sigma_{FF}, D \sim f_{\text{MV}} \left(\text{vec}(\tilde{A}_F), \Sigma_{FF} \otimes \frac{T}{T^*} (X'X)^{-1} \right) \quad (\text{IA.19})$$

$$\Sigma_{FF} | D \sim f_{\text{IW}}^k(S_F, T^* + N + K - k - m - 1) \quad (\text{IA.20})$$

$$\text{vec}(\Phi) | \Sigma_{RR}, D \sim f_{\text{MV}} \left(\text{vec}(\tilde{\Phi}), \Sigma_{RR} \otimes (W_0'W_0 + W'W)^{-1} \right) \quad (\text{IA.21})$$

$$\Sigma_{RR} | D \sim f_{\text{IW}}^{N+K-k}(S_R, T^* - (k+1)m - 1), \quad (\text{IA.22})$$

where

$$\tilde{A}_F = \frac{T}{T^*} (X'X)^{-1} \left(X'F + T_0 [\bar{f}, \bar{f}\bar{z}']' \right) = \frac{1}{T^*} \left(T (X'X)^{-1} X'F + T_0 \hat{A}_{F_0} \right) \quad (\text{IA.23})$$

$$\tilde{\Phi} = (W_0'W_0 + W'W)^{-1} \left(W'R + W_0'W_0\hat{\Phi}_0 \right) = \frac{1}{T^*} \left(T (W'W)^{-1} W'R + T_0\hat{\Phi}_0 \right) \quad (\text{IA.24})$$

$$S_F = T^* \left(\hat{V}_f + \bar{f}\bar{f}' \right) - \frac{T}{T^*} \left(T_0[\bar{f}, \bar{f}\bar{z}'] + F'X \right) (X'X)^{-1} \left(T_0[\bar{f}, \bar{f}\bar{z}']' + X'F \right) \quad (\text{IA.25})$$

$$\begin{aligned} S_R &= R_0'Q_{W_0}R_0 + \left(R - W\tilde{\Phi} \right)' \left(R - W\tilde{\Phi} \right) + \left(\tilde{\Phi} - \hat{\Phi}_0 \right)' (W_0'W_0) \left(\tilde{\Phi} - \hat{\Phi}_0 \right) \\ &= R_0'R_0 + R'R - \tilde{\Phi}' (W_0'W_0 + W'W) \tilde{\Phi} = \frac{T^*}{T} \left(R'R - \tilde{\Phi}'W'W\tilde{\Phi} \right), \end{aligned} \quad (\text{IA.26})$$

and where $\frac{W_0'W_0}{T_0} = \frac{W'W}{T}$ and $\frac{R_0'R_0}{T_0} = \frac{R'R}{T}$.

To compute the marginal likelihood, we employ equation (7) and substitute the corresponding quantities given in equations (IA.4) to (IA.22) for the

prior, likelihood, and posterior densities. This yields

$$m(D|\mathcal{M}_C) = \pi^{-\frac{1}{2}(T)(N+K)} \times \left[\frac{|W'_0 W_0|}{|W'_0 W_0 + W'W|} \right]^{\frac{1}{2}(N+K-k)} \left[\frac{T_0}{T+T_0} \right]^{\frac{1}{2}k(m+1)} \times \left[\frac{\Gamma_{N+K-k} \left(\frac{1}{2} [T^* - (k+1)m - 1] \right)}{\Gamma_{N+K-k} \left(\frac{1}{2} [T_0 - (k+1)m - 1] \right)} \right] \left[\frac{\Gamma_k \left(\frac{1}{2} [T^* + N + K - k - m - 1] \right)}{\Gamma_k \left(\frac{1}{2} [T_0 + N + K - k - m - 1] \right)} \right] \times \left[\frac{|R'_0 Q_{W_0} R_0|^{\frac{1}{2}(T_0 - (k+1)m - 1)}}{|S_R|^{\frac{1}{2}(T^* - (k+1)m - 1)}} \right] \left[\frac{|T_0 \hat{V}_f|^{\frac{1}{2}(T_0 + N + K - k - m - 1)}}{|S_F|^{\frac{1}{2}(T^* + N + K - k - m - 1)}} \right], \quad (\text{IA.27})$$

where $\Gamma_p(\cdot)$ is the multivariate gamma function, a generalization of the gamma function. As $W'W$ is a $(1 + m + k + km) \times (1 + m + k + km)$ matrix, we obtain

$$|W'_0 W_0| = \left(\frac{T_0}{T} \right)^{(1+m+k+km)} |W'W| \quad (\text{IA.28})$$

$$|W'_0 W_0 + W'W| = \left(\frac{T^*}{T} \right)^{(1+m+k+km)} |W'W|. \quad (\text{IA.29})$$

Therefore, the marginal likelihood from equation (IA.27) is

$$m(D|\mathcal{M}_C) = \pi^{-\frac{1}{2}T(N+K)} \times \left[\frac{T_0}{T^*} \right]^{\frac{1}{2}(N+K-k)(T_0+k) + \frac{1}{2}k(m+1)} \times \left[\frac{T}{T^*} \right]^{\frac{1}{2}(N+K-k)T} \times \left[\frac{\Gamma_{N+K-k} \left(\frac{1}{2} [T^* - (k+1)m - 1] \right)}{\Gamma_{N+K-k} \left(\frac{1}{2} [T_0 - (k+1)m - 1] \right)} \right] \left[\frac{\Gamma_k \left(\frac{1}{2} [T^* + N + K - k - m - 1] \right)}{\Gamma_k \left(\frac{1}{2} [T_0 + N + K - k - m - 1] \right)} \right] \times \left[\frac{|R'R - R'F(F'F)^{-1}F'R|^{\frac{1}{2}(T_0 - (k+1)m - 1)}}{|R'R - \tilde{\Phi}'W'W\tilde{\Phi}|^{\frac{1}{2}(T^* - (k+1)m - 1)}} \right] \left[\frac{|T_0 \hat{V}_f|^{\frac{1}{2}(T_0 + N + K - k - m - 1)}}{|S_F|^{\frac{1}{2}(T^* + N + K - k - m - 1)}} \right]. \quad (\text{IA.30})$$

I.B. Marginal Likelihood for the Restricted Case

The derivation of the marginal likelihood when restricting the parameters α_0 and α_1 to zero is closely related to the unrestricted case. The likelihood of the hypothetical sample $D_0 = (R_0, F_0, X_0)$ can be rewritten as

$$\begin{aligned} \mathcal{L}(D_0|Z_0, \Sigma_{RR}, \Sigma_{FF}, \Phi^R, A_F, \mathcal{M}_R) \propto \\ |\Sigma_{RR}|^{-\frac{1}{2}T_0} \exp \left\{ -\frac{1}{2} \text{Tr} \left[\Sigma_{RR}^{-1} \left(R_0' Q_{W_0^R} R_0 + \left(\Phi^R - \hat{\Phi}_0^R \right)' W_0^{R'} W_0^R \left(\Phi^R - \hat{\Phi}_0^R \right) \right) \right] \right\} \times \\ |\Sigma_{FF}|^{-\frac{1}{2}T_0} \exp \left\{ -\frac{1}{2} \text{Tr} \left[\Sigma_{FF}^{-1} \left(F_0' Q_{X_0} F_0 + \left(A_F - \hat{A}_{F_0} \right)' X_0' X_0 \left(A_F - \hat{A}_{F_0} \right) \right) \right] \right\} \end{aligned} \quad (\text{IA.31})$$

where

$$\hat{\Phi}_0^R = \left[\hat{\beta}_0, \mathbf{0}_{(N+K-k) \times km} \right]' \quad (\text{IA.32})$$

$$W_0^R = [F_0, \Xi_0]. \quad (\text{IA.33})$$

The prior distributions for the parameters Σ_{FF} and A_F remain the same as in equations (IA.7) and (IA.8). The priors for Σ_{RR} and Φ^R are given by

$$\text{vec}(\Phi^R) | \Sigma_{RR}, D_0 \sim f_{\text{MV}} \left(\text{vec}(\hat{\Phi}_0^R), \Sigma_{RR} \otimes \left(W_0^{R'} W_0^R \right)^{-1} \right) \quad (\text{IA.34})$$

$$\Sigma_{RR} | D_0 \sim f_{\text{IW}}^{N+K-k} \left(R_0' Q_{W_0^R} R_0, T_0 - km \right). \quad (\text{IA.35})$$

Equivalent to equation (IA.18), one can replace all quantities based on the hypothetical sample in terms of moments of the observed data,

$$W_0^{R'} W_0^R = T_0 \begin{pmatrix} \hat{V}_f + \bar{f} \bar{f}' & \Delta \otimes \bar{z}' \\ \Delta' \otimes \bar{z} & \hat{V}_{\Xi} + \bar{\Xi} \bar{\Xi}' \end{pmatrix}. \quad (\text{IA.36})$$

Combining the likelihood of the observed sample $D = (R, F, Z)$ with the prior distributions based on the hypothetical sample yields the following posterior distributions for Φ^R and Σ_{RR} :

$$\text{vec}(\Phi^R) | \Sigma_{RR}, D \propto f_{\text{MV}} \left(\text{vec}(\tilde{\Phi}^R), \Sigma_{RR} \otimes (W_0^{R'} W_0^R + W^{R'} W^R)^{-1} \right) \quad (\text{IA.37})$$

$$\Sigma_{RR} | D \propto f_{\text{IW}}^{N+K-k} \left(\tilde{S}_R, T^* - km \right), \quad (\text{IA.38})$$

where

$$\begin{aligned} \tilde{\Phi}^R &= \left(W_0^{R'} W_0^R + W^{R'} W^R \right)^{-1} \left(W^{R'} R + W_0^{R'} W_0^R \hat{\Phi}_0^R \right) \\ &= \frac{1}{T^*} \left(T \left(W^{R'} W^R \right)^{-1} W^{R'} R + T_0 \hat{\Phi}_0^R \right) \end{aligned} \quad (\text{IA.39})$$

$$\begin{aligned} \tilde{S}_R &= R_0' Q_{W_0^R} R_0 + \left(R - W^R \tilde{\Phi}^R \right)' \left(R - W^R \tilde{\Phi}^R \right) + \left(\tilde{\Phi}^R - \hat{\Phi}_0^R \right)' \left(W_0^{R'} W_0^R \right) \left(\tilde{\Phi}^R - \hat{\Phi}_0^R \right) \\ &= R_0' R_0 + R' R - \tilde{\Phi}^{R'} \left(W_0^{R'} W_0^R + W^{R'} W^R \right) \tilde{\Phi}^R = \frac{T^*}{T} \left(R' R - \tilde{\Phi}^{R'} W^{R'} W^R \tilde{\Phi}^R \right). \end{aligned} \quad (\text{IA.40})$$

Thus, the marginal likelihood is

$$\begin{aligned}
m(D|\mathcal{M}_C^R) &= \pi^{-\frac{1}{2}T(N+K)} \times \left[\frac{T_0}{T^*} \right]^{\frac{1}{2}(N+K-k)(k+km)+\frac{1}{2}k(m+1)} \times \\
&\quad \left[\frac{\Gamma_{N+K-k}(\frac{1}{2}[T^*-km])}{\Gamma_{N+K-k}(\frac{1}{2}[T_0-km])} \right] \left[\frac{\Gamma_k(\frac{1}{2}[T^*+N+K-k-m-1])}{\Gamma_k(\frac{1}{2}[T_0+N+K-k-m-1])} \right] \times \\
&\quad \left[\frac{|R'_0 Q_{W_0^R} R_0|^{\frac{1}{2}(T_0-km)}}{|\tilde{S}_R|^{\frac{1}{2}(T^*-km)}} \right] \left[\frac{|T_0 \hat{V}_f|^{\frac{1}{2}(T_0+N+K-k-m-1)}}{|S_F|^{\frac{1}{2}(T^*+N+K-k-m-1)}} \right] \\
&= \pi^{-\frac{1}{2}T(N+K)} \times \left[\frac{T_0}{T^*} \right]^{\frac{1}{2}(N+K-k)(T_0+k)+\frac{1}{2}k(m+1)} \times \left[\frac{T}{T^*} \right]^{\frac{1}{2}(N+K-k)T} \times \\
&\quad \left[\frac{\Gamma_{N+K-k}(\frac{1}{2}[T^*-km])}{\Gamma_{N+K-k}(\frac{1}{2}[T_0-km])} \right] \left[\frac{\Gamma_k(\frac{1}{2}[T^*+N+K-k-m-1])}{\Gamma_k(\frac{1}{2}[T_0+N+K-k-m-1])} \right] \times \\
&\quad \left[\frac{|R'R - R'F(F'F)^{-1}F'R|^{\frac{1}{2}(T_0-km)}}{|R'R - \tilde{\Phi}^{R'}W^{R'}W^R\tilde{\Phi}^R|^{\frac{1}{2}(T^*-km)}} \right] \left[\frac{|T_0 \hat{V}_f|^{\frac{1}{2}(T_0+N+K-k-m-1)}}{|S_F|^{\frac{1}{2}(T^*+N+K-k-m-1)}} \right].
\end{aligned} \tag{IA.41}$$

II. Unconditional Asset Pricing Models

II.A. Marginal Likelihood for the Unrestricted Case

Starting from the multivariate representation of the beta pricing equations (1) and (2), in the case in which α , β , and the risk premia are time-invariant, we obtain:

$$\begin{aligned}
R &= \iota_T \alpha' + F\beta' + U_R \\
&= W\Phi + U_R
\end{aligned} \tag{IA.42}$$

$$F = \iota_T A_F + U_F, \tag{IA.43}$$

where $W = [\iota_T, F]$, $\Phi = [\alpha, \beta]'$, and $U_R \in \mathbb{R}^{T \times (N+K-k)}$ and $U_F \in \mathbb{R}^{T \times k}$ are the matrices of residuals. We assume that the error terms U_R and U_F conditional on the parameter space obey the normal distribution

$$\begin{pmatrix} u_{r,t} \\ u_{f,t} \end{pmatrix} \sim \text{i.i.d. } f_{\text{MV}}(0, \Psi), \text{ where } \Psi = \begin{pmatrix} \Sigma_{RR} & 0 \\ 0 & \Sigma_{FF} \end{pmatrix}. \quad (\text{IA.44})$$

Note that the residuals U_R and U_F in equations (IA.42) and (IA.43) and the corresponding covariance matrices Σ_{RR} and Σ_{FF} in equation (IA.44) are different from their counterparts in the section above. To ease notation, we leave these quantities unchanged.

Based on the multivariate representation of the beta pricing equations, the likelihood of the hypothetical sample $D_0 = (R_0, F_0)$, denoted by the subscript 0, can be written as

$$\begin{aligned} \mathcal{L}(D_0 | \Sigma_{RR}, \Sigma_{FF}, \Phi, A_F, \mathcal{M}) &\propto \\ &|\Sigma_{RR}|^{-\frac{1}{2}T_0} \exp \left\{ -\frac{1}{2} \text{Tr} \left[\Sigma_{RR}^{-1} \left(R_0' Q_{W_0} R_0 + (\Phi - \hat{\Phi}_0)' W_0' W_0 (\Phi - \hat{\Phi}_0) \right) \right] \right\} \times \\ &|\Sigma_{FF}|^{-\frac{1}{2}T_0} \exp \left\{ -\frac{1}{2} \text{Tr} \left[\Sigma_{FF}^{-1} \left(F_0' Q_{\iota_{T_0}} F_0 + (A_F - \hat{A}_F)' T_0 (A_F - \hat{A}_F) \right) \right] \right\}, \end{aligned} \quad (\text{IA.45})$$

where

$$\hat{\Phi}_0 = [\mathbf{0}_{(N+K-k) \times 1}, \hat{\beta}_0]' \quad (\text{IA.46})$$

$$\hat{A}_{F_0} = \hat{f}'. \quad (\text{IA.47})$$

Note that the hypothetical sample is exclusively weighted against mispricing since the moments are already fixed.

Combining the likelihood of the hypothetical sample with the uninformative prior $\pi(\Sigma_{RR}, \Sigma_{FF}, \Phi, A_F) \propto |\Psi|^{-\frac{N+K+1}{2}}$, the prior distributions for the parameters $\Sigma_{RR}, \Sigma_{FF}, \Phi$, and A_F are given by

$$\text{vec}(A_F) | \Sigma_{FF}, D_0 \sim f_{\text{MV}} \left(\text{vec}(\hat{A}_{F_0}), \frac{1}{T_0} \Sigma_{FF} \right) \quad (\text{IA.48})$$

$$\Sigma_{FF} | D_0 \sim f_{\text{IW}}^k(F_0' Q_{\iota_{T_0}} F_0, T_0 + N + K - k - 1) \quad (\text{IA.49})$$

$$\text{vec}(\Phi) | \Sigma_{RR}, D_0 \sim f_{\text{MV}} \left(\text{vec}(\hat{\Phi}_0), \Sigma_{RR} \otimes (W_0' W_0)^{-1} \right) \quad (\text{IA.50})$$

$$\Sigma_{RR} | D_0 \sim f_{\text{IW}}^{N+K-k}(R_0' Q_{W_0} R_0, T_0 - 1). \quad (\text{IA.51})$$

All quantities based on the hypothetical sample in equations (IA.48) to (IA.51) can be expressed in terms of quantities observed from the actual sample. Specifically, note that

$$\begin{aligned} F_0' Q_{\iota_{T_0}} F_0 &= \left(F_0 - \iota_{T_0} \hat{A}_{F_0} \right)' \left(F_0 - \iota_{T_0} \hat{A}_{F_0} \right) \\ &= (F_0 - \iota_{T_0} \bar{f}')' (F_0 - \iota_{T_0} \bar{f}') = T_0 \hat{V}_f \end{aligned} \quad (\text{IA.52})$$

$$\begin{aligned} R_0' Q_{W_0} R_0 &= \left(R_0 - W_0 \hat{\Phi}_0 \right)' \left(R_0 - W_0 \hat{\Phi}_0 \right) \\ &= \frac{T_0}{T} \left(R' R - \hat{\Phi}_0' W' W \hat{\Phi}_0 \right) = \frac{T_0}{T} \left(R' R - \hat{\beta}_0' F' F \hat{\beta}_0 \right), \end{aligned} \quad (\text{IA.53})$$

where $\frac{W_0' W_0}{T_0} = \frac{W' W}{T}$ and $\frac{R_0' R_0}{T_0} = \frac{R' R}{T}$.

Combining the likelihood of the observed sample $D = (R, F)$ with the prior distributions based on the hypothetical sample (equations (IA.48) to

(IA.51)) yields the following posterior distribution:

$$\text{vec}(A_F) | \Sigma_{FF}, D \sim f_{\text{MV}} \left(\text{vec}(\tilde{A}_F), \frac{1}{T^*} \Sigma_{FF} \right) \quad (\text{IA.54})$$

$$\Sigma_{FF} | D \sim f_{\text{IW}}^k(S_F, T^* + N + K - k - 1) \quad (\text{IA.55})$$

$$\text{vec}(\Phi) | \Sigma_{RR}, D \sim f_{\text{MV}} \left(\text{vec}(\tilde{\Phi}), \Sigma_{RR} \otimes (W_0' W_0 + W' W)^{-1} \right) \quad (\text{IA.56})$$

$$\Sigma_{RR} | D \sim f_{\text{IW}}^{N+K-k}(S_R, T^* - 1), \quad (\text{IA.57})$$

where $T^* = T_0 + T$, $\tilde{A}_F = \bar{f}'$, and

$$\tilde{\Phi} = (W_0' W_0 + W' W)^{-1} (W' R + W_0' W_0 \hat{\Phi}_0) = \frac{1}{T^*} (T (W' W)^{-1} W' R + T_0 \hat{\Phi}_0) \quad (\text{IA.58})$$

$$S_F = T^* \hat{V}_f \quad (\text{IA.59})$$

$$\begin{aligned} S_R &= R_0' Q_{W_0} R_0 + (R - W \tilde{\Phi})' (R - W \tilde{\Phi}) + (\tilde{\Phi} - \hat{\Phi}_0)' (W_0' W_0) (\tilde{\Phi} - \hat{\Phi}_0) \\ &= R_0' R_0 + R' R - \tilde{\Phi}' (W_0' W_0 + W' W) \tilde{\Phi} = \frac{T^*}{T} (R' R - \tilde{\Phi}' W' W \tilde{\Phi}). \end{aligned} \quad (\text{IA.60})$$

To compute the marginal likelihood, we employ equation (7) and substitute the corresponding quantities given in equations (IA.45) to (IA.57) for the prior, likelihood, and posterior densities. This yields

$$\begin{aligned} m(D | \mathcal{M}_U) &= \pi^{-\frac{1}{2}(T)(N+K)} \times \left[\frac{|W_0' W_0|}{|W_0' W_0 + W' W|} \right]^{\frac{1}{2}(N+K-k)} \left[\frac{T_0}{T + T_0} \right]^{\frac{1}{2}k} \times \\ &\quad \left[\frac{\Gamma_{N+K-k}(\frac{1}{2}[T^* - 1])}{\Gamma_{N+K-k}(\frac{1}{2}[T_0 - 1])} \right] \left[\frac{\Gamma_k(\frac{1}{2}[T^* + N + K - k - 1])}{\Gamma_k(\frac{1}{2}[T_0 + N + K - k - 1])} \right] \times \\ &\quad \left[\frac{|R_0' Q_{W_0} R_0|^{\frac{1}{2}(T_0-1)}}{|S_R|^{\frac{1}{2}(T^*-1)}} \right] \left[\frac{|T_0 \hat{V}_f|^{\frac{1}{2}(T_0+N-1)}}{|S_F|^{\frac{1}{2}(T^*+N-1)}} \right]. \end{aligned} \quad (\text{IA.61})$$

As $W'W$ is a $(1+k) \times (1+k)$ matrix, we obtain

$$|W'_0 W_0| = \left(\frac{T_0}{T}\right)^{(1+k)} |W'W| \quad (\text{IA.62})$$

$$|W'_0 W_0 + W'W| = \left(\frac{T^*}{T}\right)^{(1+k)} |W'W|. \quad (\text{IA.63})$$

Thus, the marginal likelihood from equation (IA.61) is

$$\begin{aligned} m(D|\mathcal{M}_U) = & \pi^{-\frac{1}{2}T(N+K)} \times \left[\frac{T_0}{T^*}\right]^{\frac{1}{2}(N+K-k)(T_0+k)+\frac{1}{2}k} \times \left[\frac{T}{T^*}\right]^{\frac{1}{2}(N+K-k)T} \times \\ & \left[\frac{\Gamma_{N+K-k}\left(\frac{1}{2}[T^*-1]\right)}{\Gamma_{N+K-k}\left(\frac{1}{2}[T_0-1]\right)}\right] \left[\frac{\Gamma_k\left(\frac{1}{2}[T^*+N+K-k-1]\right)}{\Gamma_k\left(\frac{1}{2}[T_0+N+K-k-1]\right)}\right] \times \\ & \left[\frac{|R'R - R'F(F'F)^{-1}F'R|^{\frac{1}{2}(T_0-1)}}{|R'R - \tilde{\Phi}'W'W\tilde{\Phi}|^{\frac{1}{2}(T^*-1)}}\right] \left[\frac{|T_0\hat{V}_f|^{\frac{1}{2}(T_0+N-1)}}{|T^*\hat{V}_f|^{\frac{1}{2}(T^*+N-1)}}\right]. \end{aligned} \quad (\text{IA.64})$$

Note that for the unconditional and unrestricted case, equation (IA.64) is identical to the conditional unrestricted case in equation (IA.30) when the number of predictors in the latter equation is zeroed out.

II.B. Marginal Likelihood for the Restricted Case

The derivation of the marginal likelihood when restricting α to zero is closely related to the unrestricted case. The likelihood of the hypothetical

sample $D_0 = (R_0, F_0)$ can be rewritten as

$$\begin{aligned} \mathcal{L}(D_0 | \Sigma_{RR}, \Sigma_{FF}, \Phi^R, A_F, \mathcal{M}_R) \propto \\ |\Sigma_{RR}|^{-\frac{1}{2}T_0} \exp \left\{ -\frac{1}{2} \text{Tr} \left[\Sigma_{RR}^{-1} \left(R_0' Q_{W_0^R} R_0 + \left(\Phi^R - \hat{\Phi}_0^R \right)' W_0^{R'} W_0^R \left(\Phi^R - \hat{\Phi}_0^R \right) \right) \right] \right\} \times \\ |\Sigma_{FF}|^{-\frac{1}{2}T_0} \exp \left\{ -\frac{1}{2} \text{Tr} \left[\Sigma_{FF}^{-1} \left(F_0' Q_{\nu_{T_0}} F_0 + \left(A_F - \hat{A}_F \right)' T_0 \left(A_F - \hat{A}_F \right) \right) \right] \right\}, \end{aligned} \quad (\text{IA.65})$$

where $\hat{\Phi}_0^R = \hat{\beta}_0$ and $W_0^R = F_0$.

The prior distributions for the parameters Σ_{FF} and A_F remain the same as in equations (IA.48) and (IA.49). The priors for Σ_{RR} and Φ^R are given by

$$\text{vec}(\Phi^R) | \Sigma_{RR}, D_0 \sim f_{\text{MV}} \left(\text{vec}(\hat{\Phi}_0^R), \Sigma_{RR} \otimes (W_0^{R'} W_0^R)^{-1} \right) \quad (\text{IA.66})$$

$$\Sigma_{RR} | D_0 \sim f_{\text{IW}}^{N+K-k} \left(R_0' Q_{W_0^R} R_0, T_0 \right). \quad (\text{IA.67})$$

Combining the likelihood of the observed sample $D = (R, F)$ with the prior distributions based on the hypothetical sample yields the following posterior distributions for Φ^R and Σ_{RR} :

$$\text{vec}(\Phi^R) | \Sigma_{RR}, D \propto f_{\text{MV}} \left(\text{vec}(\tilde{\Phi}^R), \Sigma_{RR} \otimes (W_0^{R'} W_0^R + W^{R'} W^R)^{-1} \right) \quad (\text{IA.68})$$

$$\Sigma_{RR} | D \propto f_{\text{IW}}^{N+K-k} \left(\tilde{S}_R, T^* \right), \quad (\text{IA.69})$$

where

$$\tilde{\Phi}^R = \left(W_0^{R'} W_0^R + W^{R'} W^R \right)^{-1} \left(W^{R'} R + W_0^{R'} W_0^R \hat{\Phi}_0^R \right) = \hat{\Phi}_0^R = \hat{\beta}_0 \quad (\text{IA.70})$$

$$\begin{aligned} \tilde{S}_R &= R_0' Q_{W_0^R} R_0 + \left(R - W^R \tilde{\Phi}^R \right)' \left(R - W^R \tilde{\Phi}^R \right) + \left(\tilde{\Phi}^R - \hat{\Phi}_0^R \right)' \left(W_0^{R'} W_0^R \right) \left(\tilde{\Phi}^R - \hat{\Phi}_0^R \right) \\ &= \frac{T^*}{T_0} R_0' Q_{W_0^R} R_0. \end{aligned} \quad (\text{IA.71})$$

The marginal likelihood takes the form

$$\begin{aligned} m(D | \mathcal{M}_U^R) &= \pi^{-\frac{1}{2}T(N+K)} \times \left[\frac{T_0}{T^*} \right]^{\frac{1}{2}(N+K-k)k + \frac{1}{2}k} \times \\ &\quad \left[\frac{\Gamma_{N+K-k} \left(\frac{1}{2}T^* \right)}{\Gamma_{N+K-k} \left(\frac{1}{2}T_0 \right)} \right] \left[\frac{\Gamma_k \left(\frac{1}{2} [T^* + N + K - k - 1] \right)}{\Gamma_k \left(\frac{1}{2} [T_0 + N + K - k - 1] \right)} \right] \times \\ &\quad \left[\frac{|R_0' Q_{W_0^R} R_0|^{\frac{1}{2}T_0}}{|\tilde{S}_R|^{\frac{1}{2}T^*}} \right] \left[\frac{|T_0 \hat{V}_f|^{\frac{1}{2}(T_0+N+K-k-1)}}{|S_F|^{\frac{1}{2}(T^*+N+K-k-1)}} \right] \\ &= \pi^{-\frac{1}{2}T(N+K)} \times \left[\frac{T}{T^*} \right]^{\frac{1}{2}(N+K-k)(T_0+k) + \frac{1}{2}k} \times \left[\frac{T_0}{T^*} \right]^{\frac{1}{2}(N+K-k)T} \times \\ &\quad \left[\frac{\Gamma_{N+K-k} \left(\frac{1}{2}T^* \right)}{\Gamma_{N+K-k} \left(\frac{1}{2}T_0 \right)} \right] \left[\frac{\Gamma_k \left(\frac{1}{2} [T^* + N + K - k - 1] \right)}{\Gamma_k \left(\frac{1}{2} [T_0 + N + K - k - 1] \right)} \right] \times \\ &\quad \left[\frac{|R' R - R' F (F' F)^{-1} F' R|^{\frac{1}{2}T_0}}{|R' R - \tilde{\Phi}^{R'} W^{R'} W^R \tilde{\Phi}^R|^{\frac{1}{2}T^*}} \right] \left[\frac{|T_0 \hat{V}_f|^{\frac{1}{2}(T_0+N+K-k-1)}}{|S_F|^{\frac{1}{2}(T^*+N+K-k-1)}} \right]. \quad (\text{IA.72}) \end{aligned}$$

Note that for the marginal likelihood for the unconditional and restricted case in equation (IA.72) is equal to the conditional and restricted case in equation (IA.41) when the number of predictors is zeroed out.

III. Summary of the Marginal Likelihoods

The general formula for calculating the marginal likelihood is given by

$$m(D|\mathcal{M}) = \pi^{-\frac{1}{2}(T)(N+K)} \times \left[\frac{T_0}{T^*} \right]^{\frac{Q_R}{2}} \left[\frac{T_0}{T^*} \right]^{\frac{Q_F}{2}} \times \\ \left[\frac{\Gamma_{N+K-k} \left(\frac{1}{2} [T^* + \nu_R] \right)}{\Gamma_{N+K-k} \left(\frac{1}{2} [T_0 + \nu_R] \right)} \right] \left[\frac{\Gamma_k \left(\frac{1}{2} [T^* + \nu_F] \right)}{\Gamma_k \left(\frac{1}{2} [T_0 + \nu_F] \right)} \right] \times \\ \left[\frac{|R'_0 Q_{W_0} R_0|^{\frac{1}{2}(T_0 + \nu_R)}}{|S_R|^{\frac{1}{2}(T^* + \nu_R)}} \right] \left[\frac{|T_0 \hat{V}_f|^{\frac{1}{2}(T_0 + \nu_F)}}{|S_F|^{\frac{1}{2}(T^* + \nu_F)}} \right]. \quad (\text{IA.73})$$

The parameters Q_R , Q_F , ν_R , and ν_F are determined based on the model specifications in Panel A of Table I. The parameter values for each model are specified in Panel B of Table I. For models $\mathbb{M}_1 - \mathbb{M}_2$, we obtain

$$S_F = T^* \hat{V}_f \quad (\text{IA.74})$$

and for models $\mathbb{M}_3 - \mathbb{M}_4$,

$$S_F = T^* \left(\hat{V}_f + \bar{f} \bar{f}' \right) - \frac{T}{T^*} \left(T_0 [\bar{f}, \bar{f} \bar{z}'] + F' X \right) (X' X)^{-1} \left(T_0 [\bar{f}, \bar{f} \bar{z}']' + X' F \right). \quad (\text{IA.75})$$

For all models,

$$R'_0 Q_{W_0} R_0 = \frac{T_0}{T} \left(R' R - \hat{\beta}'_0 F' F \hat{\beta}_0 \right) \quad (\text{IA.76})$$

$$S_R = \frac{T^*}{T} \left(R' R - \tilde{\Phi}' W' W \tilde{\Phi} \right), \quad (\text{IA.77})$$

where R , F , S_R , S_F , and $\hat{V}_f$ vary according to the exact model specification and depend on the subgroup of the included/excluded factors and predictors.

IV. Invariance of the Marginal Likelihood to Linear Transformation of the Predictors

In this section, we show that given the following data-generating process

$$r_{t+1} = \beta_0 f_{t+1} + \beta_1 z_t f_{t+1} + u_{r,t+1}, \quad (\text{IA.78})$$

the marginal likelihood given in equation (IA.73) is invariant under a linear transformation of z_t and, in particular, standardization. To show this, we prove in the following that a linear transformation of z_t does not change S_R and S_F in equation (IA.73).

We start with the term S_R . Without loss of generality and for ease of notation, we assume that z_t and f_{t+1} in equation (IA.78) are scalars. We rewrite equation (IA.78) in matrix notation as

$$\begin{aligned} R &= F\beta'_0 + \Xi\beta'_1 + U_R \\ &= W\Phi + U_R, \end{aligned} \quad (\text{IA.79})$$

where $R = [r_1, \dots, r_T]' \in \mathbb{R}^{T \times N}$, $F = [f_1, \dots, f_T]' \in \mathbb{R}^T$, $\Xi = [\xi_1, \dots, \xi_T]' \in \mathbb{R}^T$ is defined through $\xi_t = z_{t-1}f_t$, $W = [F, \Xi]$, and $\Phi = [\beta_0, \beta_1]'$. $U_R = [u_{r,1}, \dots, u_{r,T}]' \in \mathbb{R}^{T \times N}$ is the matrix of residuals. The covariance matrix of the residuals is

$$S_R = R' \left(I_T - W(W'W)^{-1}W' \right) R, \quad (\text{IA.80})$$

where

$$(W'W)^{-1} = \frac{1}{T(\bar{f}^2\bar{\xi}^2 - \overline{f\xi^2})} \begin{bmatrix} \bar{\xi}^2 & -\overline{f\xi} \\ -\overline{f\xi} & \bar{f}^2 \end{bmatrix}. \quad (\text{IA.81})$$

We have

$$\begin{aligned} S_R &= R' \left(I - W (W'W)^{-1} W' \right) R, \\ &= R' \left(I - \frac{1}{T(\bar{f}^2\bar{\xi}^2 - \overline{f\xi^2})} \begin{bmatrix} F & \Xi \end{bmatrix} \begin{bmatrix} \bar{\xi}^2 & -\overline{f\xi} \\ -\overline{f\xi} & \bar{f}^2 \end{bmatrix} \begin{bmatrix} F' \\ \Xi' \end{bmatrix} \right) R, \\ &= R' \left(I - \frac{[\bar{\xi}^2 F' F - 2\overline{f\xi}\Xi F' + \bar{f}^2\Xi\Xi']}{T(\bar{f}^2\bar{\xi}^2 - \overline{f\xi^2})} \right) R. \end{aligned} \quad (\text{IA.82})$$

To set the stage, we start with a shift in z_t . In particular, let $\tilde{z}_t = z_t - a$, where a is a constant. Then

$$\tilde{\xi}_t = \tilde{z}_{t-1} f_t = \xi - a f \quad (\text{IA.83})$$

$$\tilde{\bar{\xi}} = \bar{\xi} - a \bar{f} \quad (\text{IA.84})$$

$$\overline{f\tilde{\xi}} = \overline{f\xi} - a \bar{f}^2 \quad (\text{IA.85})$$

$$\overline{\tilde{\xi}^2} = \bar{\xi}^2 - 2a\overline{f\xi} + a^2 \bar{f}^2. \quad (\text{IA.86})$$

For the transformed system, it follows that

$$\widetilde{S}_R = R' \left(I - \frac{[\tilde{\xi}^2 F' F - 2\overline{f\tilde{\xi}}\tilde{\Xi} F' + \bar{f}^2\tilde{\Xi}\tilde{\Xi}']}{T(\bar{f}^2\tilde{\xi}^2 - \overline{f\tilde{\xi}^2})} \right) R, \quad (\text{IA.87})$$

where $\tilde{\Xi} = \Xi - aF$. Substituting equations (IA.84), (IA.85), and (IA.86) into equation (IA.87), we obtain

$$\widetilde{S}_R = R' \left(I - \frac{[(\bar{\xi}^2 - 2a\bar{f}\bar{\xi} + a^2\bar{f}^2)F'F - 2(\bar{f}\bar{\xi} - a\bar{f}^2)(\Xi - aF)F' + \bar{f}^2(\Xi - aF)(\Xi' - aF')]}{T(\bar{f}^2(\bar{\xi}^2 - 2a\bar{f}\bar{\xi} + a^2\bar{f}^2) - (\bar{f}\bar{\xi} - a\bar{f}^2)^2)} \right) R. \quad (\text{IA.88})$$

After some algebra and the cancellation of common terms, equation (IA.88) equals equation (IA.82).

Second, consider a scale transformation of z_t , namely $\tilde{z}_t = cz_t$ where c is a scalar. The β_1 coefficients in equation (IA.78) change to $\tilde{\beta}_1 = \frac{1}{c}\beta_1$ and the covariance matrix of the residuals, S_R , remains unchanged. This completes the proof for S_R .

The proof for S_F derives from the above proof for S_R . It is a reduction of the data generating process in equation (IA.78) that is achieved by setting $f_t \equiv 1$ and changing the notation r_t to f_t . This completes the proof that a linear transformation of the predictors z_t has no effect on the marginal likelihood in equation (IA.73).

V. Derivation of T_0

We derive T_0 for two scenarios: (i) both α and β are constant, namely, both $\alpha_1 = 0$ and $\beta_1 = 0$ in equation (3), and (ii) both α and β are time varying.

V.A. α and β are Constant

When $\alpha = \alpha_0$ and $\beta = \beta_0$, the matrix $W'_0 W_0$ takes the form

$$W'_0 W_0 = T_0 \begin{pmatrix} 1 & \bar{f}' \\ \bar{f} & \hat{V}_f + \bar{f} \bar{f}' \end{pmatrix} \quad (\text{IA.89})$$

and

$$(W'_0 W_0)^{-1} = \frac{1}{T_0} \begin{pmatrix} 1 + \bar{f}' \hat{V}_f^{-1} \bar{f} & -\bar{f}' \hat{V}_f^{-1} \\ -\hat{V}_f^{-1} \bar{f} & \hat{V}_f^{-1} \end{pmatrix}. \quad (\text{IA.90})$$

Therefore,

$$\begin{aligned} \text{Var}(\alpha | \Sigma_{RR}, D_0) &= \text{Var}(\alpha_0 | \Sigma_{RR}, D_0) \\ &= \frac{\Sigma_{RR}}{T_0} \left(1 + \bar{f}' \hat{V}_f^{-1} \bar{f} \right) = \frac{\Sigma_{RR}}{T_0} (1 + SR_{max}^2), \end{aligned} \quad (\text{IA.91})$$

where SR_{max} is the Sharpe ratio of the tangency portfolio constructed by the factors.

Equating the variance of α in the hypothetical sample (equation (IA.91)) to the prior's variance in equation (15), and substituting η from equation (18), we obtain

$$T_0 = \frac{(N + K - k) (1 + SR_{max}^2)}{(\tau^2 - 1) SR^2(Mkt)}, \quad (\text{IA.92})$$

where N is the number of test assets, $K - k$ is the number of redundant factors, and $SR^2(Mkt)$ is the squared Sharpe ratio of the market.

This exact relation assigns our prior specification to the mispricing uncertainty interpretation. This prior is economically sound. In particular, when the admissible squared Sharpe ratio rises, the prior is more strongly weighted against mispricing and thereby tames the squared Sharpe ratio (based on the formulation in equation (16)). In addition, as τ increases, we permit more arbitrage opportunities that translate into generating a higher Sharpe ratio.

V.B. α and β are Time Varying

We first partition the matrix $(W_0'W_0)^{-1}$ into four blocks,

$$(W_0'W_0)^{-1} = \frac{1}{T_0} \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}, \quad (\text{IA.93})$$

where B_{11} is a $(1+m) \times (1+m)$ matrix, B_{21} is a $(1+m) \times (k+km)$ matrix, $B_{12} = B_{21}'$ and B_{22} is a $(k+km) \times (k+km)$ matrix. Specifically,

$$B_{11} = \begin{pmatrix} 1 + \bar{z}'\hat{V}_z^{-1}\bar{z} + \bar{f}'\hat{V}_f^{-1}\bar{f} + \bar{z}'\hat{V}_z^{-1}\bar{z} \times \bar{f}'\hat{V}_f^{-1}\bar{f} & -\bar{z}'\hat{V}_z^{-1} \times (1 + \bar{f}'\hat{V}_f^{-1}\bar{f}) \\ -\hat{V}_z^{-1}\bar{z} \times (1 + \bar{f}'\hat{V}_f^{-1}\bar{f}) & \hat{V}_z^{-1} \times (1 + \bar{f}'\hat{V}_f^{-1}\bar{f}) \end{pmatrix}, \quad (\text{IA.94})$$

$$B_{21} = - \begin{pmatrix} (1 + \bar{z}'\hat{V}_z^{-1}\bar{z}) \times \hat{V}_f^{-1}\bar{f} & -\hat{V}_f^{-1}\bar{f} \otimes \bar{z}'\hat{V}_z^{-1} \\ -\hat{V}_f^{-1}\bar{f} \otimes \hat{V}_z^{-1}\bar{z} & \hat{V}_f^{-1}\bar{f} \otimes \hat{V}_z^{-1} \end{pmatrix} \quad (\text{IA.95})$$

$$B_{22} = \begin{pmatrix} \hat{V}_f^{-1} \times (1 + \bar{z}' \hat{V}_z^{-1} \bar{z}) & -\hat{V}_f^{-1} \otimes \bar{z}' \hat{V}_z^{-1} \\ -\hat{V}_f^{-1} \otimes \hat{V}_z^{-1} \bar{z} & \hat{V}_f^{-1} \otimes V_z^{-1} \end{pmatrix}. \quad (\text{IA.96})$$

It follows that the unconditional variance of total mispricing (sum of fixed and time varying) is then equal to

$$\begin{aligned} \text{Var}(\alpha | \Sigma_{RR}, D_0) &= \text{Var}(\alpha_0 + \alpha'_1 z | \Sigma_{RR}, D_0) \\ &= \frac{\Sigma_{RR}}{T_0} \text{Tr} \left[B_{11} \begin{pmatrix} 1 & \bar{z}' \\ \bar{z} & \hat{V}_z + \bar{z} \bar{z}' \end{pmatrix} \right] \\ &= \frac{\Sigma_{RR}}{T_0} \text{Tr} \left[\begin{pmatrix} 1 + \bar{f}' \hat{V}_f^{-1} \bar{f} & \dots \\ \dots & I_{m \times m} (1 + \bar{f}' \hat{V}_f^{-1} \bar{f}) \end{pmatrix} \right] \\ &= \frac{\Sigma_{RR}}{T_0} (1 + \bar{f}' \hat{V}_f^{-1} \bar{f} + m(1 + \bar{f}' \hat{V}_f^{-1} \bar{f})) \\ &= \frac{\Sigma_{RR}}{T_0} (1 + SR_{max}^2 + m(1 + SR_{max}^2)), \end{aligned} \quad (\text{IA.97})$$

where Tr denotes the trace operator, and SR_{max} is the Sharpe ratio of the tangency portfolio constructed by the factors, and m is the number of predictors in the model.

Equating the unconditional variance of α in the hypothetical sample (equation (IA.97)) with the prior's variance in equation (15) and with the formulation of η from equation (18), we obtain

$$T_0 = \frac{(N + K - k) (1 + SR_{max}^2 + m(1 + SR_{max}^2))}{(\tau^2 - 1) SR^2(Mkt)}, \quad (\text{IA.98})$$

where N is the number of test assets, $K - k$ is the number of redundant factors, and $SR^2(Mkt)$ is the squared Sharpe ratio of the market.

VI. Data and Variables

Table IA.I. Variable Definitions

Variable	Abbreviation	Definition
Panel A: Asset Pricing Factors		
Market	MKT	The excess return on the value-weighted CRSP market index over the one-month Treasury bill rate (Sharpe (1964), Lintner (1965)).
Size	SMB	The small minus big firm return premium (Fama and French (1993)).
Value	HML	The high book-to-market minus the low book-to-market return premium (Fama and French (1993)).
Profitability	RMW	The robust minus the weak return premium (Fama and French (2015)).
Investment	CMA	The conservative minus the aggressive return premium (Fama and French (2015)).
Momentum	MOM	The winner minus the loser return premium (Carhart (1997)).
Post-earnings Announcement Drift	PEAD	The positive earnings surprise minus negative earnings surprise premium (Daniel et al. (2020)).
Financing	FIN	The low-issuance minus the high-issuance return premium (Daniel et al. (2020)).
Quality-minus-junk	QMJ	The quality minus the junk return premium (Asness et al. (2019)).
Betting-against-beta	BAB	The low-beta minus the high-beta return premium (Frazzini and Pedersen (2014)).
Management	MGMT	The underpriced minus the overpriced return premium based on six anomaly variables related to firms' management, including net stock issues, composite equity issues, accruals, net operating assets, asset growth, and investment to assets (Stambaugh and Yuan (2017)).
Performance	PERF	The underpriced minus the overpriced return premium based on five anomaly variables related to firms' performance, including distress, O-score, momentum, gross profitability, and return on assets (Stambaugh and Yuan (2017)).
Liquidity	LIQ	The high-liquidity-beta minus the low-liquidity-beta return premium (Pástor and Stambaugh (2003)).
Intermediary Capital	ICR	The value-weighted equity return for the primary dealer sector (He et al. (2017)).
Panel B: Macro Predictors		
Dividend Price Ratio	dp	The difference between the log of dividends and the log of prices, where dividends are 12-month moving sums of dividends paid on the S&P 500 index, and prices are monthly averages of daily closing prices (Campbell and Shiller (1988), Campbell and Yogo (2006)).
Dividend Yield	dy	The difference between the log of dividends and the log of lagged prices (Ball (1978)).
Earnings Price Ratio	ep	The difference between the log of earnings and the log of prices, where earnings are 12-month moving sums of earnings on the S&P 500 index (Campbell and Shiller (1988)).
Dividend Payout Ratio	de	The difference between the log of dividends and the log of earnings (Lamont (1998)).
Stock Variance	svar	The sum of squared daily returns on the S&P 500 index (Guo (2006)).
Book-to-market Ratio	bm	The ratio of book value to market value for the Dow Jones Industrial Average (Kothari and Shanken (1997)).
Net Equity Expansion	ntis	The ratio of 12-month moving sums of net issues by NYSE listed stocks divided by the total end-of-year market capitalization of NYSE stocks (Campbell et al. (2008)).
Treasury Bills	tbl	The <i>3-Month Treasury Bill: Secondary Market Rate</i> from the economic research database at the Federal Reserve Bank at St. Louis (Campbell (1987)).
Long Term Yield	lty	The long-term government bond yield from Ibbotson's Stocks, Bonds, Bills and Inflation Yearbook (Welch and Goyal (2008)).
Long Term Rate of Returns	ltr	The long-term government bond returns from Ibbotson's Stocks, Bonds, Bills and Inflation Yearbook (Welch and Goyal (2008)).
Term Spread	tms	The difference between the long term yield on government bonds and the Treasury bill (Campbell (1987)).
Default Yield Spread	dfy	The difference between BAA and AAA-rated corporate bond yields (Fama and French (1989)).
Inflation	infl	The <i>Consumer Price Index (All Urban Consumers)</i> from the Bureau of Labor Statistics (Campbell and Vuolteenaho (2004)).

Table IA.II. Summary Statistics

This table presents summary statistics for the data used in the paper during the period from June 1977 to December 2016. Panel A reports the mean, median, and standard deviation of monthly factor returns and the correlations between them. We also report the monthly CAPM α and its corresponding t -statistics. Panel B reports the means, medians, standard deviations, and AR(1) coefficients of the monthly macro predictors. Table IA.I provides detailed definitions for each variable.

Panel A: Summary Statistics for Factors														
	MKT	SMB	HML	RMW	CMA	MOM	PEAD	FIN	QMJ	BAB	MGMT	PERF	LIQ	ICR
Mean	0.637	0.219	0.289	0.363	0.290	0.625	0.576	0.715	0.466	0.952	0.572	0.710	0.443	1.132
Median	1.070	0.110	0.120	0.330	0.160	0.770	0.621	0.621	0.457	1.129	0.531	0.648	0.380	1.480
Std.Dev.	4.441	2.912	2.895	2.352	1.972	4.458	1.890	3.910	2.363	3.633	2.827	3.987	3.411	6.680
CAPM α		0.121	0.406	0.464	0.401	0.689	0.596	0.990	0.650	1.039	0.779	0.869	0.448	0.357
t -statistics		0.921	3.145	4.458	4.752	3.346	6.806	6.264	7.070	6.245	6.902	4.882	2.830	1.958
MKT	1.000	0.233	-0.281	-0.301	-0.390	-0.101	-0.074	-0.492	-0.546	-0.168	-0.511	-0.277	-0.010	0.810
SMB		1.000	-0.148	-0.406	-0.079	0.049	0.065	-0.417	-0.441	-0.016	-0.331	-0.101	-0.035	0.108
HML			1.000	0.214	0.690	-0.221	-0.175	0.637	0.021	0.385	0.715	-0.299	0.022	0.026
RMW				1.000	0.120	0.115	-0.076	0.653	0.752	0.377	0.344	0.431	-0.043	-0.226
CMA					1.000	-0.011	-0.034	0.599	0.120	0.352	0.767	-0.019	0.002	-0.216
MOM						1.000	0.462	0.039	0.273	0.224	0.003	0.728	0.024	-0.256
PEAD							1.000	-0.067	0.146	0.067	-0.023	0.396	0.049	-0.170
FIN								1.000	0.586	0.495	0.789	0.184	0.007	-0.293
QMJ									1.000	0.248	0.399	0.643	-0.024	-0.528
BAB										1.000	0.369	0.187	0.069	-0.047
MGMT											1.000	0.026	-0.015	-0.261
PERF												1.000	0.016	-0.483
LIQ													1.000	-0.022
ICR														1.000

Panel B: Summary Statistics for Macro Predictors													
	dp	dy	ep	de	svar	bm	ntis	tbl	lty	ltr	tms	dfy	infl
Mean	-3.670	-3.663	-2.878	-0.792	0.003	0.444	0.006	0.047	0.069	0.007	0.022	0.011	0.003
Median	-3.790	-3.785	-2.898	-0.853	0.001	0.337	0.009	0.049	0.065	0.008	0.024	0.009	0.003
Std.Dev.	0.439	0.440	0.483	0.355	0.005	0.273	0.020	0.036	0.030	0.032	0.015	0.005	0.004
AR(1)	0.994	0.994	0.988	0.985	0.456	0.993	0.980	0.993	0.996	0.049	0.948	0.963	0.625

VII. Predictive Regressions

VII.A. Posterior Probabilities for Predictive Regressions

The analysis of predictive regressions is motivated by taking expectations from both sides of equation (3) conditional on z_t and the underlying model parameters, as well as using the factor generating process in equation (2). In particular, expected returns are given by

$$E[r_{t+1}|z_t, \theta] = \left(\alpha_0 + \beta_0 \alpha_f \right) + \left(\alpha_1 + \beta_0 a_F + \beta_1 (\alpha_f \otimes I_M) \right) z_t + \beta_1 (a_F \otimes I_M) (z_t \otimes z_t). \quad (\text{IA.99})$$

Thus, when both factor loadings and risk premia are time varying, expected stock returns depend on macro predictors through their levels, squared values, and interactions.

Hence, a general formulation for predictive regressions should allow for nonlinear relations between future returns and predictors. To illustrate, focusing on the model that retains all M macro predictors, the predictive regression is formulated as

$$y_{t+1} = B_0 + B_1 z_t + B_2 \text{vech}(z_t z_t') + \varepsilon_{t+1}, \quad (\text{IA.100})$$

where the operator $\text{vech}(z_t z_t')$ vectorizes the entries below and on the main diagonal of the matrix $z_t z_t'$ and is a $\frac{1}{2}(M+1) \times M$ vector that contains all interaction terms of the predictive variables, $B = [B_0, B_1, B_2]$ is a $(N+K) \times (1 + \frac{1}{2}M(M+3))$ matrix of the regression intercept and slope coefficients,

and ε_{t+1} is assumed to obey $\varepsilon_{t+1} \sim N(0, \Sigma)$.

The set of retained predictors is unique for every predictive regression. A candidate set of predictors is defined through a vector of binary variables of length M , indicating the inclusion or exclusion of a variable. The number of retained predictors is denoted by m , which ranges between zero and M . In one extreme case, all predictors are excluded, and returns are independently and identically distributed (IID). The polar extreme is the all-inclusive specification that retains all M predictors.

We compute the marginal likelihoods for the collection of predictive regressions on the basis of three scenarios. The first excludes nonlinearities by setting $B_2 = 0$. The second allows for squared values and interactions. The third is a combination of the first two scenarios. We provide more details in the empirical section that follows. We can then assess the inclusion probability for both individual predictors and nonlinearities.

It is convenient to reformulate the data-generating process in equation (IA.100) using matrix notation

$$Y = XB + U, \tag{IA.101}$$

where $X = [x_0, x_1, \dots, x_{T-1}]'$, $Y = [y_1, y_2, \dots, y_T]'$, $U = [\varepsilon_1, \dots, \varepsilon_T]'$, and $x_t = [1, z_t']'$ if interaction terms are omitted or $x_t = [1, z_t', \text{vech}(z_t z_t')]'$ if interaction terms are included.

The marginal likelihood computation for the predictive regression is based on observing a hypothetical sample of length T_0 that is weighted against predictability. In that sample, the slope coefficients in the predictive regressions

are centered around zero. The size of the hypothetical prior is set as a fixed number, that is, 50, multiplied by the number of parameters underlying each of the predictive models, following Kandel and Stambaugh (1996).

We show below that the log marginal likelihood of any predictive regression is given by

$$\begin{aligned} \ln [m(D|\mathcal{M})] &= -\frac{T(N+K)}{2} \ln(\pi) + \frac{T_0 - m - 1}{2} \ln |T_0 \hat{V}_y| - \frac{T^* - m - 1}{2} \ln |\tilde{S}| \\ &\quad - \sum_{i=1}^{N+K} \ln \left\{ \Gamma \left(\frac{T_0 - m - i}{2} \right) \right\} + \sum_{i=1}^{N+K} \ln \left\{ \Gamma \left(\frac{T^* - m - i}{2} \right) \right\} \\ &\quad - \frac{(N+K)(m+1)}{2} \ln \left(\frac{T^*}{T_0} \right), \end{aligned} \quad (\text{IA.102})$$

where

$$\tilde{S} = T^* \left(\hat{V}_y + \bar{y}\bar{y}' \right) - \frac{T}{T^*} (T_0 \bar{y}\bar{x}' + Y'X) (X'X)^{-1} (T_0 \bar{x}\bar{y}' + X'Y), \quad (\text{IA.103})$$

and where m is the number of retained predictors, $\bar{x} = \frac{1}{T} \sum_{t=1}^T x_t$, $T^* = T + T_0$, $\Gamma(\phi)$ denotes the Gamma function evaluated at ϕ , and $|A|$ is the determinant of matrix A . For the IID model, it follows that $\tilde{S}_{iid} = T^* \hat{V}_y$, where $\hat{V}_y$ is defined in equation (9).

VII.B. Marginal Likelihood of Predictive Regressions

The marginal likelihood derivation for predictive regressions, per equation (IA.100), is based on Avramov (2002), with essential modifications to account for nonlinearities in the predictors as well as interactions. First, it is convenient to reformulate the data generating process in equation

(IA.100) in matrix notation $Y = XB + U$, where $X = [x_0, x_1, \dots, x_{T-1}]'$, $Y = [y_1, y_2, \dots, y_T]'$, $U = [\varepsilon_1, \dots, \varepsilon_T]'$, and $x_t = [1, z_t']'$ if interaction terms are omitted or $x_t = [1, z_t', \text{vech}(z_t z_t')]'$ if interaction terms are included. The set of selected predictors differs across models, but we refrain from model-specific subscripts to keep the notation simple.

We conduct Bayesian inference for the primary sample based on the joint posterior distribution of B and Σ from the hypothetical sample equipped with a noninformative prior. Multiplying the likelihood of the hypothetical sample with the noninformative prior $\pi(B, \Sigma) \propto |\Sigma|^{-\frac{N+K+1}{2}}$ yields the informative prior, formulated as

$$\pi(B, \Sigma | D_0) \propto |\Sigma|^{-\frac{T_0+N+K+1}{2}} \exp \left(-\frac{1}{2} \text{Tr} \left[(S_0 + (B - B_0)' X_0' X_0 (B - B_0)) \Sigma^{-1} \right] \right), \quad (\text{IA.104})$$

where

$$S_0 = (Y_0 - X_0 B_0)' (Y_0 - X_0 B_0) = T_0 \hat{V}_y \quad (\text{IA.105})$$

$$B_0 = \begin{bmatrix} \bar{y}' \\ 0 \end{bmatrix} \quad (\text{IA.106})$$

$$(X_0' X_0) = \frac{T_0}{T} (X' X) = T_0 \overline{xx'}, \quad (\text{IA.107})$$

and $\overline{xx'}$ is the sample mean of xx' .

Standard results imply that the priors for B and Σ can be expressed as

$$\text{vec}(B) | \Sigma, D_0 \sim f_{\text{MV}} \left(\text{vec}(B_0), \Sigma \otimes (X_0' X_0)^{-1} \right), \quad (\text{IA.108})$$

$$\Sigma | D_0 \sim f_{\text{IW}}^{N+K} (S_0, T_0 - m - 1), \quad (\text{IA.109})$$

where $\text{vec}(\cdot)$ denotes the vector formed by stacking the successive transformed rows of a matrix, m is the number of retained predictors, MV denotes the multivariate normal distribution, IW denotes the inverted Wishart density, and $N + K$ denotes the degrees of freedom of the inverted Wishart density. We refer the reader to Zellner (1971) for further technical details.

Combining the likelihood of the actual sample D and the priors in equations (IA.108) and (IA.109) yields the posterior density

$$\text{vec}(B) | \Sigma, D \sim N \left(\text{vec}(\tilde{B}), \Sigma \otimes (X_0' X_0 + X' X)^{-1} \right) \quad (\text{IA.110})$$

$$\Sigma | D \sim f_{\text{IW}}^{N+K} \left(\tilde{S}, T^* - m - 1 \right), \quad (\text{IA.111})$$

where $T^* = T + T_0$,

$$\tilde{B} = \frac{T}{T^*} (X' X)^{-1} (T_0 \bar{x} \bar{y}' + X' Y), \quad (\text{IA.112})$$

$$\tilde{S} = T^* \left(\hat{V}_y + \bar{y} \bar{y}' \right) - \frac{T}{T^*} (T_0 \bar{y} \bar{x}' + Y' X) (X' X)^{-1} (T_0 \bar{x} \bar{y}' + X' Y). \quad (\text{IA.113})$$

The log marginal likelihood for the predictive regression setup formulated

in equation (IA.100) then takes the form

$$\begin{aligned}
\ln [m(D|\mathcal{M})] &= -\frac{T(N+K)}{2} \ln(\pi) + \frac{T_0 - m - 1}{2} \ln |T_0 \hat{V}_y| - \frac{T^* - m - 1}{2} \ln |\tilde{S}| \\
&- \sum_{i=1}^{N+K} \ln \left\{ \Gamma \left(\frac{T_0 - m - i}{2} \right) \right\} + \sum_{i=1}^{N+K} \ln \left\{ \Gamma \left(\frac{T^* - m - i}{2} \right) \right\} \\
&- \frac{(N+K)(m+1)}{2} \ln \left(\frac{T^*}{T_0} \right), \tag{IA.114}
\end{aligned}$$

where $\Gamma(\phi)$ is the Gamma function evaluated at ϕ . $|x|$ is the determinant of x .

VII.C. Empirical Results: Posterior Probabilities of Predictors in Predictive Regressions

Table IA.III. Posterior Probabilities of Predictors in Predictive Regressions

This table presents the cumulative posterior probabilities for the macro predictors in predictive regressions, given as $\mathcal{A}'\mathcal{P}$, where $\mathcal{A}$ is a matrix representing all forecasting models by their unique combinations of zeros and ones, and $\mathcal{P}$ is a vector including posterior probabilities for all models. We consider three scenarios described in equation (IA.100): (i) no interaction ($B_1 \neq 0$ and $B_2 = 0$), (ii) with interaction ($B_1 \neq 0$ and $B_2 \neq 0$), and (iii) combined, including both (i) and (ii). Table IA.I provides the detailed definitions for each variable.

	No Interaction	With Interaction	Combined
dp	0.15	0.35	0.35
dy	1.00	1.00	1.00
ep	0.85	0.65	0.65
de	0.85	0.65	0.65
svar	0.95	1.00	1.00
bm	0.72	0.04	0.04
ntis	0.74	0.69	0.69
tbl	0.76	0.36	0.36
lty	0.24	0.63	0.63
ltr	0.85	0.31	0.31
tms	0.67	0.24	0.24
dfy	0.72	1.00	1.00
infl	0.24	0.24	0.24

VIII. Bayesian Predictive Moments

Starting from equation (2) and denoting the posterior means of the parameters θ with $\tilde{\theta}$, the mean and variance of the predicted factor returns for model $\mathcal{M}_l$ are

$$E[f_{t+1}|D_t, \mathcal{M}_l] = \tilde{\alpha}_{l,f} + \tilde{\alpha}_{l,F} z_t = \hat{f}_{l,t+1}, \quad (\text{IA.115})$$

$$\begin{aligned} \text{Var}[f_{t+1}|D_t, \mathcal{M}_l] &= [I_k \otimes x'_t] \left[\tilde{\Sigma}_{FF} \otimes \frac{t}{t^*} (X'X)^{-1} \right] [I_k \otimes x'_t]' + \tilde{\Sigma}_{FF} \\ &= \frac{1}{t^*} [I_k \otimes x'_t] \left[\tilde{\Sigma}_{FF} \otimes \begin{pmatrix} 1 + \bar{z}' \hat{V}_z^{-1} \bar{z} & -\bar{z}' \hat{V}_z^{-1} \\ -\hat{V}_z^{-1} \bar{z} & \hat{V}_z^{-1} \end{pmatrix} \right] [I_k \otimes x_t] + \tilde{\Sigma}_{FF} \\ &= \frac{1}{t^*} \tilde{\Sigma}_{FF} \otimes x'_t \begin{pmatrix} 1 + \bar{z}' \hat{V}_z^{-1} \bar{z} & -\bar{z}' \hat{V}_z^{-1} \\ -\hat{V}_z^{-1} \bar{z} & \hat{V}_z^{-1} \end{pmatrix} x_t + \tilde{\Sigma}_{FF} \\ &= \tilde{\Sigma}_{FF} \left(1 + \frac{(1 + (\bar{z} - z_t)' \hat{V}_z^{-1} (\bar{z} - z_t))}{t^*} \right) \\ &= \tilde{\Sigma}_{FF} (1 + \delta_{t,t^*}), \end{aligned} \quad (\text{IA.116})$$

where $x_t = [1, z'_t]'$, t is the sample size, t_0 is the hypothetical sample size corresponding to the sample t , $t^* = t + t_0$, $\delta_{t,t^*} = \frac{(1 + (\bar{z} - z_t)' \hat{V}_z^{-1} (\bar{z} - z_t))}{t^*}$, $\tilde{\Sigma}_{FF} = \frac{S_F}{t^* + N + K - 2k - m - 2}$, and the matrix S_F is defined in equation (IA.25).

From equation (3), the mean and variance of the predicted future returns

for the test assets for model $\mathcal{M}_l$ are

$$E[r_{t+1}|D_t, \mathcal{M}_l] = \tilde{\alpha}_{l,0} + \tilde{\alpha}_{l,1}z_t + \left(\tilde{\beta}_{l,0} + \tilde{\beta}_{l,1}(I_k \otimes z_t) \right) (\tilde{\alpha}_{l,f} + \tilde{a}_{l,F}z_t), \quad (\text{IA.117})$$

$$\begin{aligned} \text{Var}[r_{t+1}|D_t, \mathcal{M}_l] &= [I_{N+K-k} \otimes \hat{w}'_{t+1}] \left[\tilde{\Sigma}_{RR} \otimes \frac{t}{t^*} (W'W)^{-1} \right] [I_{N+K-k} \otimes \hat{w}'_{t+1}]' \\ &\quad + (1 + \delta_{t,t^*}) \left(\tilde{\beta}_{l,0} + \tilde{\beta}_{l,1}(I_k \otimes z_t) \right) \tilde{\Sigma}_{FF} \left(\tilde{\beta}_{l,0} + \tilde{\beta}_{l,1}(I_k \otimes z_t) \right)' + \tilde{\Sigma}_{RR} \\ &= \tilde{\Sigma}_{RR} \left(1 + \frac{t}{t^*} \hat{w}'_{t+1} (W'W)^{-1} \hat{w}_{t+1} \right) \\ &\quad + (1 + \delta_{t,t^*}) \left(\tilde{\beta}_{l,0} + \tilde{\beta}_{l,1}(I_k \otimes z_t) \right) \tilde{\Sigma}_{FF} \left(\tilde{\beta}_{l,0} + \tilde{\beta}_{l,1}(I_k \otimes z_t) \right)', \end{aligned} \quad (\text{IA.118})$$

where $\hat{w}_{t+1} = [1, z'_t, \hat{f}'_{t+1}, \hat{f}'_{t+1} \otimes z'_t]$, $\tilde{\Sigma}_{RR} = \frac{\tilde{S}_R}{t^* - (k+1)m - N - K + k - 2}$, and the matrix $\tilde{S}_R$ is defined in equation (IA.26).

In the restricted setup, both $\alpha_0 = 0$ and $\alpha_1 = 0$ in equation (3). The mean and variance of the predicted future returns for the test assets for model $\mathcal{M}_l$ are

$$E[r_{t+1}|D_t, \mathcal{M}_l] = \left(\tilde{\beta}_{l,0} + \tilde{\beta}_{l,1}(I_k \otimes z_t) \right) (\tilde{\alpha}_{l,f} + \tilde{a}_{l,F}z_t), \quad (\text{IA.119})$$

$$\begin{aligned} \text{Var}[r_{t+1}|D_t, \mathcal{M}_l] &= [I_{N+K-k} \otimes \hat{w}^{R'}_{t+1}] \left[\tilde{\Sigma}_{RR} \otimes \frac{t}{t^*} (W'W)^{-1} \right] [I_{N+K-k} \otimes \hat{w}^{R'}_{t+1}]' \\ &\quad + (1 + \delta_{t,t^*}) \left(\tilde{\beta}_{l,0} + \tilde{\beta}_{l,1}(I_k \otimes z_t) \right) \tilde{\Sigma}_{FF} \left(\tilde{\beta}_{l,0} + \tilde{\beta}_{l,1}(I_k \otimes z_t) \right)' + \tilde{\Sigma}_{RR} \\ &= \tilde{\Sigma}_{RR} \left(1 + \frac{t}{t^*} \hat{w}^{R'}_{t+1} (W^{R'}W^R)^{-1} \hat{w}^{R'}_{t+1} \right) \\ &\quad + (1 + \delta_{t,t^*}) \left(\tilde{\beta}_{l,0} + \tilde{\beta}_{l,1}(I_k \otimes z_t) \right) \tilde{\Sigma}_{FF} \left(\tilde{\beta}_{l,0} + \tilde{\beta}_{l,1}(I_k \otimes z_t) \right)', \end{aligned} \quad (\text{IA.120})$$

where $\hat{w}_{t+1}^R = [\hat{f}'_{t+1}, \hat{f}'_{t+1} \otimes z'_t]'$, $\tilde{\Sigma}_{RR} = \frac{\tilde{S}_R}{t^* - km - N - K + k - 1}$, and the matrix $\tilde{S}_R$ is defined in equation (IA.40).

$$E[r_{t+1}|D_t, \mathcal{M}_l] = \tilde{\alpha}_{l,t}(z_t) + \tilde{\beta}_l(z_t)(\tilde{\alpha}_{l,f} + \tilde{a}_{l,F}z_t) \quad (\text{IA.121})$$

$$\text{Var}[r_{t+1}|D_t, \mathcal{M}_l] = A_{t+1} + (1 + \delta_{t+1})\tilde{\beta}(z_t)\tilde{\Sigma}_{FF}\tilde{\beta}(z_t)' \quad (\text{IA.122})$$

where A_{t+1} is an $(N + K - k) \times (N + K - k)$ diagonal matrix whose i -th entry is $\tilde{\Psi}(1 + \delta_{0,t+1})$ with diagonal matrix $\tilde{\Psi} := \frac{1}{t-m-k-km-3}S_{R,ii}$ (matrix S_R is defined in equation (IA.26)), $\delta_{t+1} = \frac{1}{t+1} \left(1 + (\bar{z} - z_t)' \hat{V}_z^{-1} (\bar{z} - z_t)\right)$, and $\tilde{\Sigma}_{FF} = \frac{t}{t-m-2k-3}\tilde{\Sigma}_{FF}$. The scalar $\delta_{0,t+1}$ is defined as

$$\begin{aligned} \delta_{0,t} &= x'_t \tilde{W}^{11} x_t + 2x'_t \left[\tilde{W}^{12} + \tilde{W}^{13} (I_k \otimes z_t) \right] \tilde{A}'_F x_t \\ &\quad + \text{tr} \left[\tilde{A}'_F x_t x'_t \tilde{A}_F \tilde{W} \right] + (1 + \delta_{t+1}) \text{tr} \left[\tilde{\Sigma}_{FF} \tilde{W} \right] \end{aligned} \quad (\text{IA.123})$$

$$\begin{aligned} \tilde{W} &= \tilde{W}^{22} + \tilde{W}^{23} (I_k \otimes z_t) + (I_k \otimes z'_t) \tilde{W}^{32} + (I_k \otimes z'_t)' \tilde{W}^{33} (I_k \otimes z'_t). \end{aligned} \quad (\text{IA.124})$$

The matrix $\tilde{W}^{mn}$ corresponds to the mn -th block of the partition $(W'_0 W_0 + W' W)^{-1}$, where W_0 is defined in equation (IA.18) and $W = [1, z'_{t-1}; f'_t; f'_t \otimes z'_{t-1}]$.

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